

HEALTHCONNECT WEEK 6 PROJECT SUMMARY

Data Analytics Track

AnalystLab Africa - Experience Lab

1. WHAT I PLANNED TO ACCOMPLISH

- Deepen Week 5 analysis with segment investigation
- Validate KPIs using statistical testing
- Integrate with Data Science track
- Improve dashboard with new findings
- Develop evidence-based recommendations

2. WHAT I COMPLETED

- Advanced segmentation analysis (appointment type, age, gender)
- Statistical validation of KPIs (Chi-Square, T-Test, Correlation)
- Cross-track integration with Data Science track
- Updated dashboard with new findings
- 5 evidence-based recommendations with priority levels

3. WHAT I IMPROVED FROM WEEK 5

Week 5 Output	Week 6 Improvement
Initial EDA	Advanced/decision-support analysis
Calculated KPIs	Validated/refined KPIs
Initial dashboard	Improved dashboard with findings
Basic insights	Evidence-based recommendations
Identified collaboration	Actually integrated with another track

4. WHAT I INTEGRATED

- Shared findings with Data Science track
- Received feature importance feedback from their model
- Aligned on key predictors for modelling
- Validated my analytical conclusions against their model

5. WHICH TRACK(S) I COLLABORATED WITH

Primary: Data Science Track

Secondary: Project Management Track (coordination)

6. WHAT WAS EXCHANGED

I Provided to Data Science:

- Validated KPIs (No-Show Rate: 48%)
- Segment analysis (High-risk: 18.5%)
- Feature recommendations (previous_no_shows, booking_lead_days)
- Statistical validation results

I Received from Data Science:

- Feature importance from baseline model
- Confirmation of key predictors
- Model performance insights

7. WHAT CHANGED AS A RESULT

- Better understanding of which features matter most
- Validated insights for recommendations
- Improved dashboard with model insights
- Alignment on key predictors for modelling
- More targeted recommendations

8. KEY FINDINGS

Finding	Value
No-Show Rate	48.0%
High-Risk Patients (2+ no-shows)	18.5%
Reminder Effectiveness	15.2% reduction
Booking Lead Days	Strong correlation
Previous No-Shows	Strongest predictor
Best Day (lowest no-shows)	Sunday

Worst Day (highest no-shows)

Monday

9. MAJOR CHALLENGES

1. Statistical test interpretation required additional research
2. Cross-track coordination timing was difficult
3. SciPy installation issues in Python environment
4. Histogram not available in Power BI (used Column Chart)

10. IMPORTANT DECISIONS

1. Excluded Cancelled appointments from No-Show validation
2. Used Chi-Square for categorical variable testing
3. Used T-Test for continuous variable comparison
4. Created binary is_no_show column for calculations
5. Used Yes/No for reminder_sent instead of TRUE/FALSE

11. REMAINING ISSUES

1. Need more data on patient satisfaction
2. Limited demographic information available
3. No external data (weather, traffic, public transport)
4. Booking lead days needs further investigation
5. Reminder channel effectiveness varies

12. CONTRIBUTION TO OVERALL HEALTHCONNECT PROJECT

- Validated analytics for model development
- Provided feature recommendations for Data Science
- Delivered evidence-based recommendations
- Created reusable analysis framework
- Established cross-track collaboration process

13. PROPOSED FOCUS FOR WEEK 7

- Final dashboard refinement
- Testing with real scenarios
- Final recommendations
- Prepare for Week 8 presentation

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